



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)
ALLEN COMMON TEST
09-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, TRAS, I(A), I, II & LIVE-I(A), I, I(E)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,D	B,C	A,B,C,D	A,C	A	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	C	D	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	4.85 to 4.95	4.20 to 4.30	35.00	1.67	64.00	12.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	2	7			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,B,C	A,B,C	A,B,C,D	A,B,C	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	C	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	6.00	0.00	10.50	8.00	3.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	9	4	5			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C	A,C	B,D	A,D	A,B,C,D	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	B	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	216.00	5.00	7.00	-1.57	8.00	8.00
SECTION-II (ii)	Q.	7	8	9			
	A.	828	124	3			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,D)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{dv}{dt} = -\frac{v^2}{u^2} \frac{du}{dt} = -8 \text{ cm/s}$$

$$m_T = -\frac{v}{u} = \frac{f}{f-u}$$

$$h_i = \frac{f}{f-u} h_0$$

$$\frac{dh_i}{dt} = \frac{fh_0}{(f-u)^2} \frac{du}{dt} = \frac{-20 \times 0.2 \times 2}{100} = \frac{-2}{25} \text{ cm/s}$$

2. Ans (B,C)

$$\frac{1}{2} CV_{\max}^2 + 0 = \frac{1}{2} CV^2 + \frac{1}{2} Li^2$$

$$360 = \frac{1}{2} CV^2 + 320 \quad \dots(i)$$

$$\frac{1}{2} CV^2 = 40 \mu\text{J}$$

$$\frac{1}{2} C V_{\max}^2 = 360$$

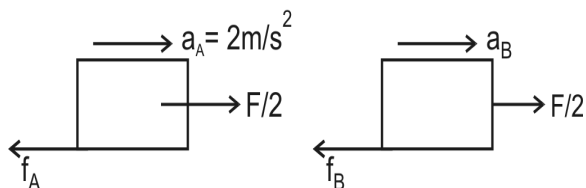
$$\frac{1}{2} \times C \times 15^2 = 360$$

$$C = 3.2 \mu\text{F}$$

Solve equation (i) to get V & it is 5V.

3. Ans (A,B,C,D)

Block B will slip relative to plank C but the plank A will not slip relative to plank C.



$$f_B = 0.1 \times 10 = 1 \text{ N}$$

$$f_A + f_B = m_C a_C \Rightarrow f_A + 1 = 1 \times 2 \Rightarrow f_A = 1 \text{ N}$$

$$\frac{F}{2} - f_A = m_A a_A \Rightarrow \frac{F}{2} - 1 = 4 \times 2$$

$$F = 18 \text{ N}$$

$$\frac{F}{2} - f_B = m_B a_B$$

$$a_B = 8 \text{ m/s}^2$$

4. Ans (A,C)

R_2 is the reaction when block B is stationary

$$R_2 = mg \cos \theta = mg \cos 30^\circ = \frac{mg\sqrt{3}}{2}$$

$$\text{given } \frac{R_1}{R_2} = \frac{4}{3}$$

$$R_1 = \frac{2mg}{\sqrt{3}}$$

If block B moves with acceleration a towards left.

$$R_1 = mg \cos 30^\circ + ma \cos 60^\circ = \frac{2mg}{\sqrt{3}}$$

$$a = \frac{g}{\sqrt{3}}$$

For A to remain stationary relative to B

$$mg \sin 30^\circ = ma \cos 30^\circ$$

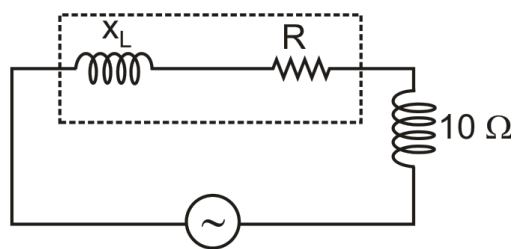
$$\frac{mg}{2} = \frac{mg}{\sqrt{3}} \cdot \frac{1}{2} = \frac{mg}{2}$$

Hence A remains stationary relative to B, if

$$\frac{R_1}{R_2} = \frac{4}{3} \text{ and for which block B moves with}$$

$$\text{acceleration } \frac{g}{\sqrt{3}}.$$

5. Ans (A)



If the current is lagging the voltage and power factor of the box is 0.5., then the box must contain a resistance in series with an inductor.

Now the circuit looks like as shown.

$$\frac{R^2}{X_L^2 + R^2} = (0.5)^2 \quad \dots(i)$$

$$\frac{(R + 10)^2}{X_L^2 + (R + 10)^2} = \left(\frac{\sqrt{3}}{2}\right)^2 \quad \dots(ii)$$

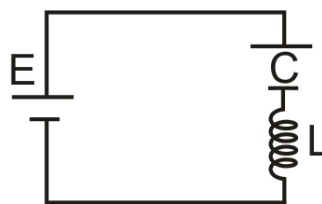
solving equation (i) & (ii)

$$R = 5\Omega \quad X_L = 5\sqrt{3}\Omega$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (C)



Charge on capacitor at any time $t = CE(1 - \cos \omega t)$

$$\omega = \frac{1}{\sqrt{LC}}$$

$$q = CE \left(1 - \cos \cdot \frac{1}{\sqrt{LC}} \sqrt{LC} \frac{\pi}{2} \right) = CE$$

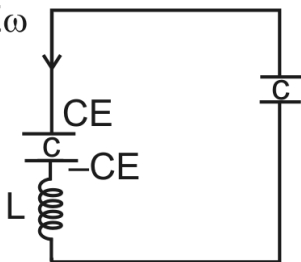
$$i = \frac{dq}{dt} = CE\omega \sin \omega t$$

$$\text{at } t = t_0$$

$$q = CE\omega$$

8. Ans (C)

$$i = CE\omega$$



At max current $\frac{di}{dt} = 0$

So $L \frac{di}{dt} = 0$

So potential difference across inductor is zero
so potential of both capacitor become same

So $\frac{1}{2}CE^2 + \frac{1}{2}Li^2 = \frac{1}{2}2CV^2 + \frac{1}{2}Li_{\max}^2$

$CE^2 + Li^2 = 2CV^2 + Li_{\max}^2$

$CE = 2CV \Rightarrow V = \frac{E}{2}$

$CE^2 + Li^2 = \frac{CE^2}{2} + Li_{\max}^2$

$CE^2 + LC^2E^2\omega^2 = \frac{CE^2}{2} + Li_{\max}^2$

$CE^2 + CE^2 = \frac{CE^2}{2} + Li_{\max}^2$

$\frac{3CE^2}{2L} = i_{\max}^2$

$i_{\max} = \sqrt{\frac{3C}{2L}}E$

9. Ans (D)

Component of electric field perpendicular to
intersecting area of plane with sphere is always

along x-axis

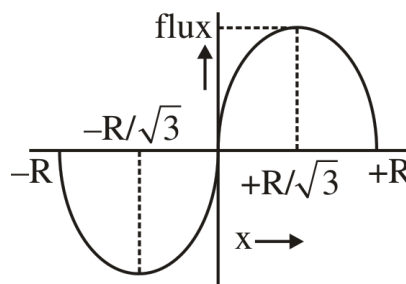
$E_x = \frac{Q}{4\pi\epsilon_0 R^3} \left(\frac{R}{2} \right) = \frac{Q}{8\pi\epsilon_0 R^2}$

Radius of intersecting area

$r = \sqrt{R^2 - \frac{R^2}{4}} = \sqrt{3} \frac{R}{2}$

$Q = E_x \pi r^2 = \frac{Q}{8\pi\epsilon_0 R^2} \times \pi \frac{3R^2}{4} = \frac{3Q}{32\epsilon_0}$

10. Ans (C)



$E_x = \frac{Qx}{4\pi\epsilon_0 R^3}$

Area = $\pi(R^2 - x^2)$

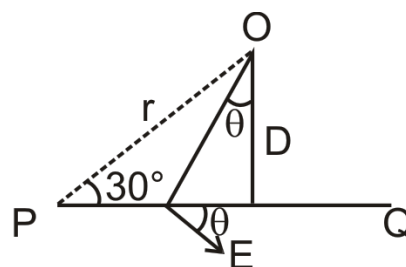
Flux = $\frac{Q}{4\epsilon_0 R^3} (R^2 x - x^3)$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (4.20 to 4.30)

O is the centroid of triangle



$E 2\pi r = \pi r^2 \frac{dB}{dt}$

$E = \frac{B_0 r}{2}$

$a_t = \frac{qE}{m} \cos\theta = \frac{qB_0 D}{2m}$

$3\ell = \frac{1}{2} \frac{qB_0 D}{2m} t^2$

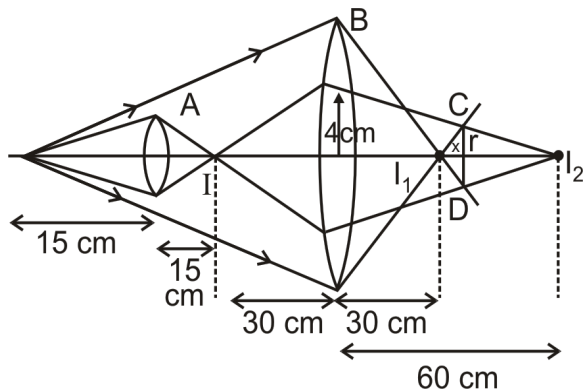
$t = \sqrt{\frac{12\ell}{\left(\frac{qB_0}{m}\right) D}}$

$\frac{D}{\ell/2} = \tan 30^\circ \Rightarrow D = \frac{\ell}{2\sqrt{3}} = 1$

$t = \sqrt{24} = 4.90 \text{ sec.}$

$v^2 = 2a_t(3\ell) = 2 \times \frac{qB_0 D}{2m} \times 3 \times 2\sqrt{3} = 4.24 \text{ m/s}$

4. Ans (1.67)



The position of Image I, formed due to due to refraction through the lens B.

$$\frac{1}{v_1} - \frac{1}{-60} = \frac{1}{20} \Rightarrow v_1 = 30 \text{ cm.}$$

The position of image I₂ formed due to double refraction through the lenses A & B

$$\frac{1}{v_1} - \frac{1}{-30} = \frac{1}{20} \Rightarrow v_2 = 60 \text{ cm.}$$

$$\text{now, } \frac{r}{x} = \frac{10}{30} \Rightarrow x = 3r$$

$$\text{also, } \frac{r}{30-x} = \frac{4}{60} \Rightarrow 15r = 30 - 3r$$

$$r = \frac{30}{18} = 5/3 = 1.67$$

$$x = 3r = 5 \text{ cm}$$

$$d = 30 + x = 35 \text{ cm}$$

6. Ans (12.00)

Let I₀ be minimum current required to flip, then

$$I_0 \ell^2 B = \frac{4mg\ell}{2}$$

$$I_0 = \frac{2mg}{\ell B}$$

$$I = \frac{4mg}{\ell B} = 2I_0$$

net torque about one edge

$$\frac{5m\ell^2}{3} \alpha = I\ell^2 B - \frac{4mg\ell}{2}$$

$$\alpha = \frac{6g}{5\ell}$$

$$a_{\text{cm}} = \frac{\alpha\ell}{2} = \frac{3g}{5}$$

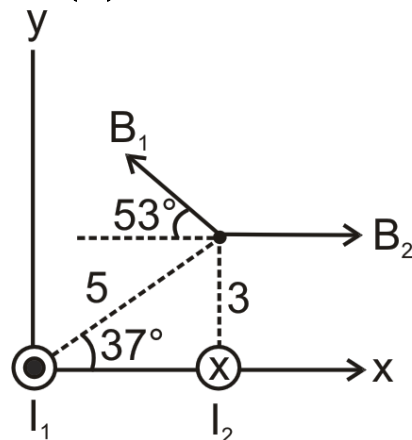
$$N - 4mg = 4m \cdot \frac{3g}{5}$$

$$N = \frac{32mg}{5}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (3)



$$B_x = 0$$

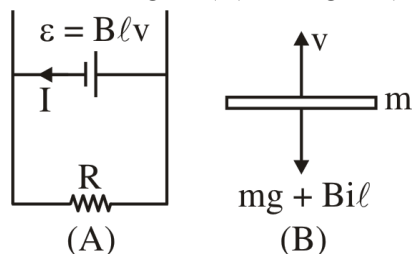
$$B_2 = B_1 \cos 53^\circ \Rightarrow B_2 = \frac{3B_1}{5}$$

$$\frac{\mu_0 I_2}{2\pi \times 3} = \frac{3}{5} \frac{\mu_0 I_1}{2\pi(5)}$$

$$\frac{I_1}{I_2} = \frac{25}{9} = 2.777$$

8. Ans (2)

Let v be the speed of rod at any time. Then the equivalent and free body diagram of rod are shown in figure (A) and figure (B) respectively.



Applying Newton's second law to rod,

$$\frac{mdv}{dt} = -(mg + Bi\ell) \quad \dots(1)$$

$$\text{where } i = \frac{Blv}{R} \quad \dots(2)$$

From equation (1) and (2)

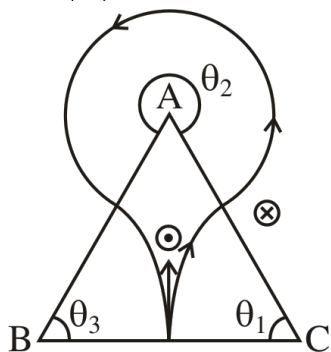
$$\frac{mdv}{dt} = - \left(mg + \frac{B^2 \ell^2 v}{R} \right)$$

integrating between proper limits we get

$$\int_0^t \frac{mdv}{mg + \frac{B^2 \ell^2 v}{R}} = \int_0^\ell -dt$$

$$t = \frac{mR}{B^2 \ell^2} \ln \frac{mg + \frac{B^2 \ell^2 u}{R}}{mg} \approx 2.1$$

9. Ans (7)



Radius of circular path

$$R = \frac{mv}{qB} = \frac{0.1 \times \pi}{0.1 \times \pi/3} = 3m$$

So it will meet side AC at mid-point of AC.

Path of charged particle is shown. So, total angle swept out by charged particle is

$$\theta_1 + \theta_2 + \theta_3 = \frac{7\pi}{3}$$

$$\text{So, time} = \frac{7\pi}{3} \times \frac{m}{qB} = \frac{7\pi}{3} \times \frac{0.1}{0.1 \times \pi/3} = 7 \text{ sec}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A,B,C,D)

Based on fajan's rule smaller the size of cation more the polarising power and larger the size of anion more the polarisation so more covalent character.

2. Ans (A,B,C)

Based on molecular orbital theory.

5. Ans (A,B,C)

Seven membered ring of azulene is electrophilic and five membered ring is nucleophilic

6. Ans (A,B,C)

In option (A) Stephen reduction

In option (B) Etard reaction

In option (C) Hydration reaction

In option (D) Benzoic acid will be the major product.

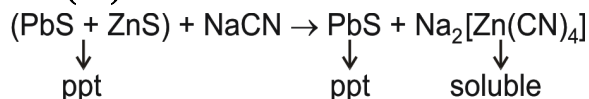
PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (C)

Factual Basis.

8. Ans (C)



PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (6.00)

The degree of polynomial in r is 3

$$\therefore \text{No of radial node} = n - \ell - 1 = 3$$

$$\therefore \text{No of angular node} = \ell = 1$$

$$\therefore n = 5$$

$$\therefore n + \ell = 5 + 1 = 6$$

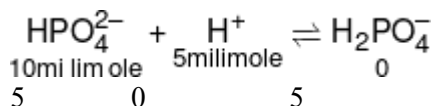
2. Ans (0.00)

Factual basis.

3. Ans (10.50)

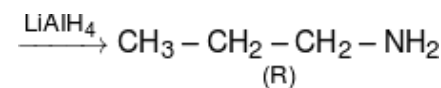
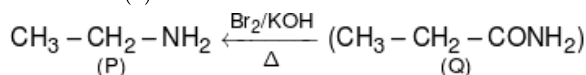
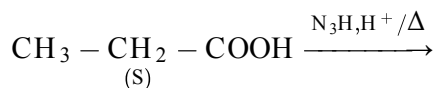
$$\text{pH} = \frac{1}{2}(\text{pK}_{a_2} + \text{pK}_{a_3}) = \frac{1}{2}(8 + 13) = 10.5$$

4. Ans (8.00)



$$\therefore \text{pH} = 8 + \log \frac{5}{5} = 8.0$$

6. Ans (3.00)



Statement 1, 3 and 4 are correct

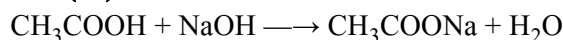
PART-2 : CHEMISTRY

SECTION-II (ii)

8. Ans (4)

Zn, Fe, Sn, Pb

9. Ans (5)



$$t = 0 \quad 2 \text{ m mol} \quad 1 \text{ m mol} \quad 1 \text{ m mol}$$

$$t = \infty \quad 1 \text{ m mol} \quad 0 \quad 2 \text{ m mol}$$

$$\begin{aligned} \text{pH} &= \text{pK}_a + \log \left(\frac{2}{1} \right) \\ &= 5 - \log 2 + \log 2 \end{aligned}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B,C)**

$$I = \int \frac{2\sqrt{x} dx}{2\sqrt{x}(1+\sqrt{x})^{2010}};$$

$$\text{Put } 1 + \sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt$$

$$= 2 \int \frac{(t-1)dt}{t^{2010}} = 2 \int (t^{-2009} - t^{-2010}) dt = 2$$

$$\left(\frac{t^{-2008}}{-2008} - \frac{t^{-2009}}{-2009} \right) + C$$

$$= \frac{2}{2009(1+\sqrt{x})^{2009}} - \frac{2}{2008(1+\sqrt{x})^{2008}} + C$$

$$\therefore \alpha = 2009, \beta = 2008$$

2. **Ans (A,C)**

$$\text{King } I(m) = \int_0^{\pi} \ln(1 - 2m \cos(\pi - x) + m^2) dx$$

$$= \int_0^{\pi} \ln(1 + 2m \cos x + m^2) dx$$

$$\text{Adding } 2I(m) = \int_0^{\pi} \ln((1+m^2)^2 - 4m^2 \cos^2 x) dx$$

$$= \int_0^{\pi} \ln(1 + 2m^2(1 - 2\cos^2 x) + m^4) dx$$

$$\Rightarrow 2I(m) = \int_0^{\pi} \ln(1 - 2m^2 \cos 2x + m^4) dx$$

$$\text{Putting } 2x = t$$

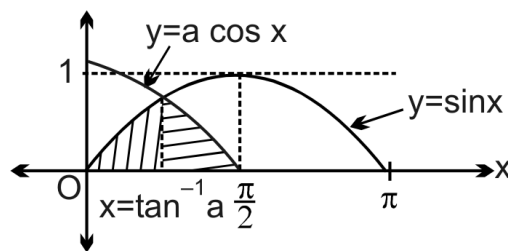
$$2I(m) = \frac{1}{2} \int_0^{2\pi} \ln(1 - 2m^2 \cos t + m^4) dt$$

$$\Rightarrow 2I(m) = \int_0^{\pi} \ln(1 - 2m^2 \cos x + m^4) dx = I(m^2)$$

$$\therefore 2I(1) = I(1) \Rightarrow I(1) = 0$$

$$\text{and } 2I(3) = I(9) \Rightarrow \frac{I(9)}{I(3)} = 2$$

3. **Ans (B,D)**



$$\text{We have } S = \int_0^{\pi} \sin x dx = 2, \text{ so } T = \frac{2}{3}$$

where $a > 0$.

$$\text{Now } T = \int_0^{\tan^{-1} a} \sin x dx + \int_{\tan^{-1} a}^{\pi/2} a \cos x dx = \frac{2}{3}$$

$$\text{i.e. } -\cos(\tan^{-1} a) + 1 + a(1 - \sin(\tan^{-1} a)) = \frac{2}{3}$$

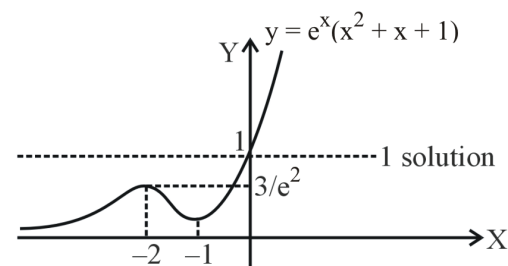
$$\text{i.e. } -\frac{1}{\sqrt{1+a^2}} + 1 + a - \frac{a^2}{\sqrt{1+a^2}} = \frac{2}{3}$$

$$\Rightarrow (a+1) - \sqrt{a^2+1} = \frac{2}{3}$$

$$\Rightarrow a + \frac{1}{3} = \sqrt{a^2+1}$$

$$\Rightarrow a = \frac{4}{3}$$

4. **Ans (A,D)**



$$\frac{dy - dx}{x^2 \left(\frac{xdy - ydx}{x^2} \right)} = \frac{d(e^{-x}) - d(x^2)}{y^2 \left(\frac{ydx - xdy}{y^2} \right)}$$

$$\Rightarrow dy - dx = d(x^2) - d(e^{-x})$$

$$\Rightarrow y - x = x^2 - e^{-x} + C$$

$$\Rightarrow y = x^2 + x - e^{-x} + C \text{ at } x = 0, y = 0, C = 1.$$

$$f(x) = x^2 + x + 1 - e^{-x}$$

$$f(x) = 0 \Rightarrow x^2 + x + 1 = e^{-x}$$

5. Ans (A,B,C,D)

$$\frac{x^2}{16} + \frac{y^2}{7} = 1 \quad \dots(1)$$

$$\Rightarrow a^2 = 16, b^2 = 7$$

$$\text{i.e. } a = 4, b = \sqrt{7}$$

$$\therefore e^2 = \frac{a^2 - b^2}{a^2} \Rightarrow e = \frac{3}{4}$$

$$\therefore \text{foci} \equiv (\pm ae, 0) = (\pm 3, 0)$$

$$\frac{x^2}{\left(\frac{144}{25}\right)} - \frac{y^2}{\left(\frac{81}{25}\right)} = 1 \quad \dots(2)$$

$$\Rightarrow a^2 = \frac{144}{25}, b^2 = \frac{81}{25}$$

$$\text{i.e. } a = \frac{12}{5}, b = \frac{9}{5}$$

$$\therefore e^2 = \frac{a^2 + b^2}{a^2}$$

$$\Rightarrow e = \frac{5}{4}$$

$$\text{foci} \equiv (\pm ae, 0) = (\pm 3, 0)$$

$$\text{solving (1) and (2) we get } y^2 = \frac{63}{25}$$

$$\Rightarrow y = \pm \frac{3\sqrt{7}}{5}$$

$$\Rightarrow x = \pm \frac{16}{5}$$

$$\text{one of the point of intersection is } \left(\frac{16}{5}, \frac{3\sqrt{7}}{5}\right)$$

$$\text{The equation of the asymptote is } \frac{x^2}{144} - \frac{y^2}{81} = 0$$

$$\text{The abscissa of P is } \frac{16}{5}$$

$$\text{Its ordinate is given by } \frac{y^2}{81} = \frac{16 \times 16}{25 \times 144}$$

$$\therefore y = \pm \frac{12}{5}$$

$$\therefore P \equiv \left(\frac{16}{5}, \frac{12}{5}\right)$$

$$\Rightarrow \left(\frac{16}{5}\right)^2 + \left(\frac{12}{5}\right)^2 = 16$$

Equation of the auxilliary circle formed on

major axis of ellipse $x^2 + y^2 = 16$, P lies on it.

6. Ans (A,C)

Any vector r in space can be written as

$$\vec{r} = \lambda \vec{a} + \mu \vec{b} + k(\vec{a} \times \vec{b})$$

take dot product with $\vec{a} \times \vec{b}$, we get $k = \frac{5}{4}$

$$\text{Let } \vec{r} \cdot \vec{a} = \lambda; \vec{r} \cdot \vec{b} = \mu$$

$$\vec{r} \cdot (3\vec{a} + 2\vec{b}) = 0$$

$$\Rightarrow 3\lambda + 2\mu = 0$$

$$\text{Given } \int_{-2\lambda}^{-\frac{4}{3}\mu} \frac{x+1}{x^2+1} dx = \int_{-2\lambda}^{2\lambda} \frac{x+1}{x^2+1} dx = 2$$

$$\int_0^{2\lambda} \frac{dx}{x^2+1} = 2 \tan^{-1}(2\lambda) = \frac{\pi}{2}$$

$$\Rightarrow \lambda = \frac{1}{2} \text{ and } \mu = -\frac{3}{4}$$

$$\therefore \vec{r} = \frac{\vec{a}}{2} - \frac{3}{4}\vec{b} + \frac{5}{4}(\vec{a} \times \vec{b})$$

$$\text{and } |\vec{r}| = \sqrt{\frac{1}{4} + \frac{9}{16} + \frac{25}{16}}$$

$$\Rightarrow |\vec{r}|^2 = \frac{19}{8}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (D)

$$\vec{a} \times \vec{c} = \vec{b} \times \vec{c} \Rightarrow (\vec{a} - \vec{b}) \times \vec{c} = \vec{0} \Rightarrow \vec{a} = \vec{b} + \mu \vec{c}$$

$$\Rightarrow ((\lambda \log_3 x) \hat{i} + 2\hat{j} + \hat{k})$$

$$= ((\log_3 x) \hat{i} + (\lambda \log_3 x) \hat{j} - \hat{k}) + (\mu \hat{i} + \mu \hat{j} + \mu \hat{k})$$

$$\lambda \log_3 x = \mu + \log_3 x; 2 = \lambda \log_3 x + \mu \text{ and } 1 = \mu - 1$$

$$\therefore \mu = 2$$

$$\lambda \log_3 x = 0 \text{ and } \log_3 x = -2$$

$$\therefore \lambda = 0$$

$$\Rightarrow \mu = 2, \log_3 x = -2, \lambda = 0$$

$$\vec{a} = 2\hat{j} + \hat{k}$$

$$\vec{b} = -2\hat{i} - \hat{k}$$

$$\vec{c} = \hat{i} + \hat{j} + \hat{k}$$

$$\frac{|\vec{b} - \vec{c}|}{|\vec{a} - \vec{c}|} = \frac{|-3\hat{i} - \hat{j} - 2\hat{k}|}{|-\hat{i} + \hat{j}|} = \sqrt{7}$$

8. Ans (B)

$$(\vec{a} \times \vec{b}) \times \vec{c} = \vec{a} + 2\vec{b}$$

$$\Rightarrow -(\vec{b} \cdot \vec{c}) \vec{a} + (\vec{a} \cdot \vec{c}) \vec{b} = \vec{a} + 2\vec{b}$$

$$\Rightarrow \vec{b} \cdot \vec{c} = -1, \vec{a} \cdot \vec{c} = 2$$

$$\Rightarrow \log_3 x + \lambda \log_3 x - 1 = -1 \quad \dots (1)$$

$$\Rightarrow \lambda \log_3 x + 2 + 1 = 2$$

$$\lambda \log_3 x = -1 \quad \dots (2)$$

$$\log_3 x = 1, \lambda = -1$$

$$\vec{a} = -\hat{i} + 2\hat{j} + \hat{k}$$

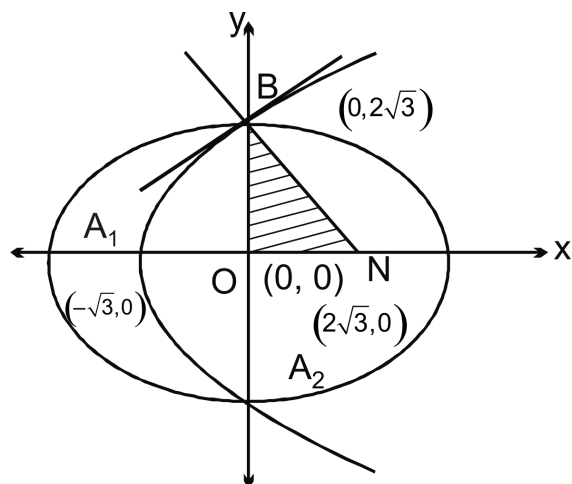
$$\vec{b} = \hat{i} - \hat{j} - \hat{k}$$

$$\vec{c} = \hat{i} + \hat{j} + \hat{k}$$

$$[\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] = [\vec{a} \quad \vec{b} \quad \vec{c}]^2 = 4$$

10. Ans (B)

Equation of parabola is $y^2 = 4\sqrt{3}(x + \sqrt{3})$



Normal to the parabola at B meets the x axis at N $(2\sqrt{3}, 0)$ and normal to the ellipse at B is the y-axis.

$$\therefore \text{Required area} = \frac{1}{2} \times 2\sqrt{3} \times 2\sqrt{3} = 6 \text{ sq-units}$$

$$\frac{A_1}{A_2} = \frac{\frac{1}{2} \cdot \pi \cdot 4 \cdot 2\sqrt{3} - \frac{2}{3} \cdot 12}{\frac{1}{2} \pi \cdot 4 \cdot 2\sqrt{3} + \frac{2}{3} \cdot 12}$$

$$= \frac{\pi\sqrt{3} - 2}{\pi\sqrt{3} + 2}$$

PART-3 : MATHEMATICS

SECTION-II (i)

2. Ans (5.00)

$$\int \left(\frac{p \cos px}{x^2} - \frac{2 \sin px}{x^3} \right) dx$$

$$= \int p \cdot \underbrace{\frac{1}{x^2}}_I \cdot \underbrace{\cos px}_{II} dx - \int \frac{2 \sin px}{x^3} dx$$

$$= p \left(\frac{1}{x^2} \cdot \frac{\sin px}{p} - \int \frac{-2}{x^3} \cdot \frac{\sin px}{p} dx \right) -$$

$$\int \frac{2 \sin px}{x^3} dx + c = \frac{\sin px}{x^2} + c = \frac{\sin px}{x} + c$$

$$\Rightarrow f(x) = \frac{\sin px}{x}$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\sin px}{x \cdot p} \cdot p = p$$

$$(1) \lim_{x \rightarrow 0} \left(\frac{p - f(x)}{x^2} \right) = \lim_{x \rightarrow 0} \left(\frac{p - \frac{\sin px}{x}}{x^2} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{px - \sin px}{p^3 x^3} \right) p^3 = \frac{p^3}{6}$$

$$\therefore \sum_{p=1}^8 \frac{p^3}{6} = \frac{1}{6} \cdot \left(\frac{8 \cdot 9}{2} \right)^2 = 216$$

$$(2) y = \left[\lambda g \left(\frac{x}{p} \right) \right]$$

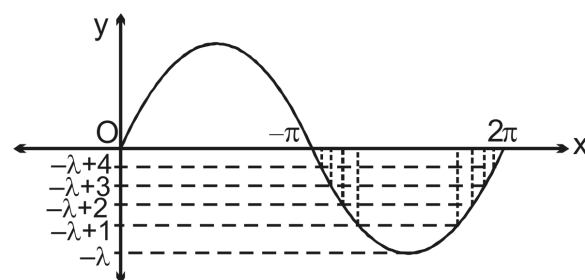
$$g(x) = x f(x) = \sin(px)$$

$$\therefore g \left(\frac{x}{p} \right) = \sin x$$

$$\Rightarrow y = [\lambda \sin x]$$

For the function to be discontinuous at exactly

8 points in $(\pi, 2\pi)$, λ must be 5.



4. Ans (-1.57)

(3) Let $g(x) = x + \sin x$

$g'(x) = 1 + \cos x$

$g'(x) \geq 0 \Rightarrow g(x)$ is $\uparrow$

For $x \in [-\pi, \pi]$, $g(x) \in [-\pi, \pi]$

$\therefore$ Integral values of $g(x)$ are $-3, -2, -1, 0, 1, 2, 3$

Hence, $f(x)$ is discontinuous at exactly 7 points.

(4) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x) dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [x + \sin x] dx$

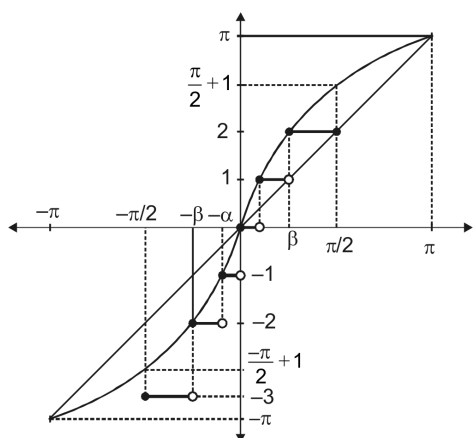
$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [x + \sin x] dx$

Applying king,

$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [-x - \sin x] dx$

$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (-1) dx = -\pi \Rightarrow I = \frac{-\pi}{2}$

Aliter :



$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} [x + \sin x] dx$

$$= \int_{-\frac{\pi}{2}}^{-\beta} -3 \cdot dx + \int_{-\beta}^{-\alpha} -2 \cdot dx + \int_{-\alpha}^0 -1 \cdot dx + \int_0^{\alpha} 0 \cdot dx + \int_{\alpha}^{\beta} 1 \cdot dx + \int_{\beta}^{\frac{\pi}{2}} 2 \cdot dx$$

$$= -3 \left(-\beta + \frac{\pi}{2} \right) - 2(-\alpha + \beta) - 1(0 + \alpha) + 0 + \beta - \alpha + 2 \left(\frac{\pi}{2} - \beta \right)$$

$$= 3\beta - \frac{3\pi}{2} + 2\alpha - 2\beta - \alpha + \beta - \alpha + \pi - 2\beta$$

$$= \frac{-\pi}{2}$$

6. Ans (8.00)

$(f'(x))^2 - f(x) f'(x) = f(x) f'(x) - f(x) f''(x)$

$f'(x) (f'(x) - f(x)) = f(x) (f'(x) - f''(x))$

$\frac{f'(x)}{f(x)} = - \left(\frac{f'(x) - f''(x)}{f(x) - f'(x)} \right)$

Integrating both sides

$\ln f(x) = - \ln (f(x) - f'(x)) + c$

$x = 0, f(0) = 1, f'(0) = 2 \Rightarrow c = 0$

$\therefore f(x) = \frac{1}{f'(x) - f(x)}$

$\Rightarrow f'(x) = \frac{f^2(x) + 1}{f(x)}$

$\Rightarrow \frac{f(x)}{f^2(x) + 1} \cdot f'(x) = 1$

$\Rightarrow \frac{1}{2} \ln (f^2(x) + 1) = x + k$

$x = 0, c = \frac{1}{2} \ln 2$

$\Rightarrow f^2(x) + 1 = 2e^{2x}$

$\Rightarrow f(x) = \sqrt{2e^{2x} - 1}$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (828)

$P(2+x) + P(2-x) = 2P(2) \dots (i)$

$x \rightarrow x-2$ in (i)

$P(x) + P(4-x) = 2P(2) \dots (ii)$

$x \rightarrow x+1$ in (i)

$P(x+3) + P(1-x) = 2P(2) \dots (iii)$

(ii) + (iii)

$20 = 4P(2)$

$\Rightarrow P(2) = 5$

now differentiate (i) w.r.t. x

$P'(2+x) - P'(2-x) = 0$

$\Rightarrow P'(2+x) = P'(2-x)$

$\therefore P'(x)$ is symmetric about $x = 2$

$P'(x) = 3x^2 + 2ax + 9$

$\Rightarrow -\frac{2a}{6} = 2$

$\Rightarrow a = -6$

$P(2) = 8 + 4a + 18 + b = 5$

$8 - 24 + 18 + b = 5$

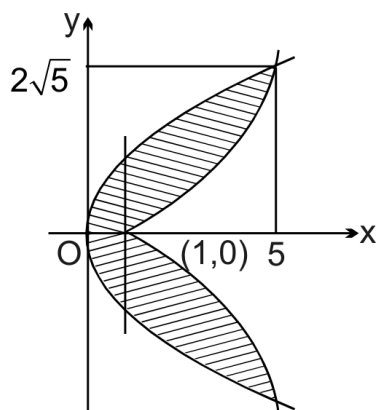
$b = 3$

$\therefore P(x) = x^3 - 6x^2 + 9x + 3$

$I = \int_{-6}^6 (x^3 - 6x^2 + 9x + 3) dx = \int_{-6}^6 (6x^2 - 3) dx$

$$= 2 \int_0^6 (6x^2 - 3) dx = 828$$

8. Ans (124)



$$y^2 = 4x ; 16y^2 = 5(x-1)^3$$

$$\text{solving, } 64x = 5(x-1)^3$$

$$\Rightarrow (x-5)(5x^2 + 10x + 1) = 0$$

$$x = 5 \text{ and } y = 2\sqrt{5}$$

$$\text{hence } A = 2 \left[\int_0^5 2\sqrt{x} \, dx - \int_1^5 \frac{\sqrt{5}}{4} (x-1)^{3/2} dx \right]$$

which on evaluating gives,

$$A = \frac{104\sqrt{5}}{15}$$

$$\therefore L = 104; M = 5; N = 15$$

$$\therefore L + M + N = 124$$

9. Ans (3)

$$\vec{a} \cdot \vec{b} = (x^2\hat{i} - 3\hat{j} + (x+3)\hat{k}) \cdot (\hat{i} + 3\hat{j} - (x-3)\hat{k})$$

$$= x^2 - 9 - (x^2 - 9) = 0$$

$$\Rightarrow \vec{a} \perp \vec{b} \text{ and } |\vec{a}| = |\vec{b}|$$

$$\vec{v}_1 \cdot \vec{v}_2 = (2\vec{a} + 3\vec{b}) \cdot (3\vec{a} - 2\vec{b})$$

$$= 6a^2 - 6b^2 - 4\vec{a} \cdot \vec{b} + 9\vec{b} \cdot \vec{a} = 0$$

$$\therefore \vec{v}_1 \wedge \vec{v}_2 = \frac{\pi}{2}$$

$$\text{Now, } I = \int_1^3 (\vec{v}_1 \wedge \vec{v}_2) \, dx = \int_1^3 \frac{\pi}{2} \cdot dx$$

$$= \frac{\pi}{2} (3-1) = \pi$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)
ALLEN COMMON TEST
09-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE-S, TRAS,I(A),I,II & LIVE-I(A),I,II(E)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	B	D	C	A	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	B	B	A	B	C	D	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	12	2	2	2	30	132	0	25	7	10

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	C	A	B	B	D	B	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	A	C	C	D	A	C	D	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2	6	917	10	3	7	2	4	16	18

PART-3 : MATHEMATICS

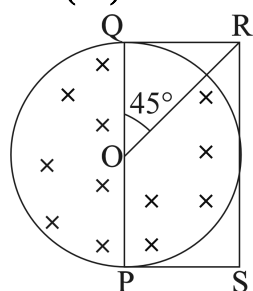
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	B	D	C	A	A	C	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	C	C	D	A	A	B	B	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	12	2010	2014	7	2	9	10	8	15	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)



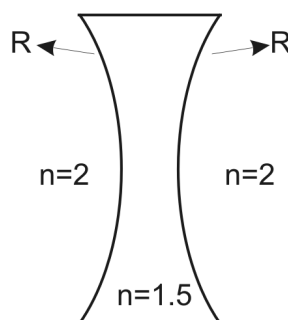
$$\phi_{OQR} = \frac{\pi R^2 B}{8} = 5t \left(\frac{\pi R^2}{8} \right)$$

$$\text{E.M.F Induced in QR} = \left| \frac{-d\phi_{PQR}}{dt} \right| = \frac{5\pi R^2}{8}$$

E.M.F Induced in PQ = 0

Because electric field lines are perpendicular to PQ

2. Ans (B)



$$\frac{1}{F} = \left[\frac{1.5}{2} - 1 \right] \left[\frac{-1}{R} - \frac{1}{R} \right]$$

$$\frac{1}{F} = \frac{1}{2R}$$

$F = 2R$ Convergent [B]

3. **Ans (C)**

$$L_{eq} = 2L$$

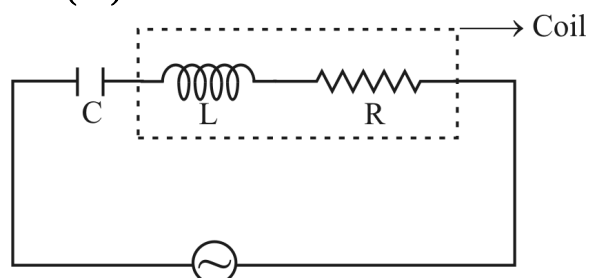
$$\tau = 2L/R$$

$$\text{Current through source 'i'} = \frac{V}{R} \left[1 - e^{-\frac{Rt}{2L}} \right]$$

Current divide in inverse ratio in parallel inductor circuit;

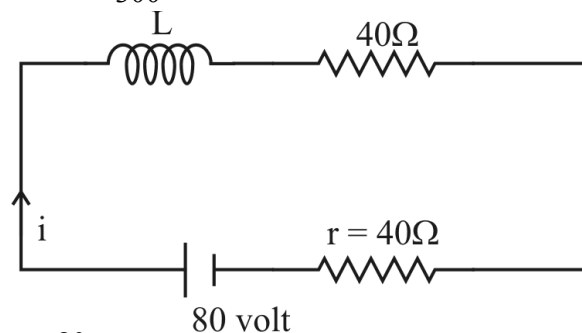
$$i_{3L} = \frac{2V}{3R} \left[1 - e^{-\frac{Rt}{2L}} \right] [C]$$

4. **Ans (B)**



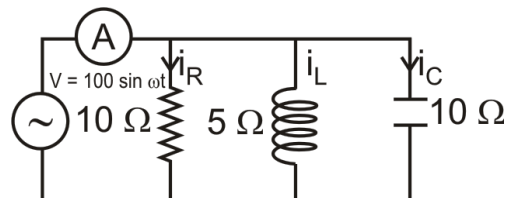
$$i_{\max} = \frac{12}{R} = 300 \times 10^{-3} \text{ A}$$

$$R = \frac{12000}{300} = 40\Omega$$



$$i = \frac{80}{80} = 1\text{A}$$

5. **Ans (B)**



$$i_R = 10 \sin \omega t$$

$$i_L = 20 \sin \left[\omega t - \frac{\pi}{2} \right]$$

$$i_C = 10 \sin \left[\omega t + \frac{\pi}{2} \right]$$

$$i = i_R + i_L + i_C$$

$$i = 10 \sqrt{2} \sin \left[\omega t - \frac{\pi}{4} \right]$$

reading of ammeter = 10 A

6. **Ans (D)**

$$-\frac{1}{f_m} = \frac{2}{30}$$

$$\frac{1}{f_m} = -\frac{1}{15}$$

Now,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f_m}$$

$$\frac{1}{v} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60}$$

$$v = 60 \text{ cm}, \text{ Real}$$

7. **Ans (C)**

$$Q_{\text{factor}} = \frac{f_0}{\text{Band width}}$$

$$= \frac{800}{850 - 750} = 8 \text{ (C)}$$

8. **Ans (A)**

$$8 = \frac{f_0}{f_e}$$

$$f_0 = 8f_e \text{ (i)}$$

$$18 = f_0 + f_e = 9f_e$$

$$f_e = 2 \text{ cm}; f_0 = 16 \text{ cm (i)}$$

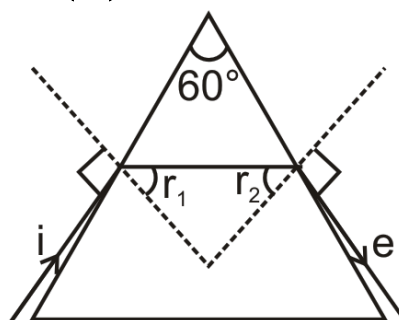
9. **Ans (B)**

$$\frac{\text{Force}}{\text{length}} = \frac{\mu_0 i_1 i_2}{2\pi d}$$

$$= \frac{\mu_0 (5)(40)}{2\pi (1)}$$

$$= \frac{100\mu_0}{\pi} \text{ (B) repulsive}$$

10. **Ans (B)**



$$1 \sin 90^\circ = 2 \sin r_1$$

$$r_1 = 30^\circ$$

$$\Rightarrow r_2 = 30^\circ$$

$$\Rightarrow e = 90^\circ$$

$$\delta = 90^\circ + 90^\circ - 60^\circ$$

$$= 120^\circ$$

11. Ans (A)

$$T = 2\pi \sqrt{\frac{I}{MB_H}}$$

$$T = 2\pi \sqrt{\frac{mx^2}{MB_H}}$$

$$T \propto \sqrt{m}$$

$$\frac{T_1^2}{T_2^2} = \frac{m_1}{m_2}$$

$$\frac{T}{T_2^2} = \frac{m}{16m}$$

$$T_2 = 4T$$

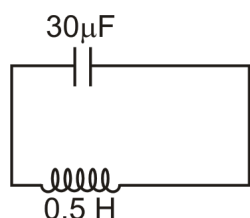
12. Ans (B)

$$\tan \theta' = \frac{\tan \theta}{\cos \alpha}$$

$$\tan \theta' = \frac{\tan 30^\circ}{\cos 30^\circ} = \frac{2}{3}$$

$$\theta' = \tan^{-1} \left(\frac{2}{3} \right)$$

13. Ans (B)



$$Q = 15 \times 10^{-5} \text{ C}$$

$$\text{voltage across capacitor} = \frac{Q}{C}$$

$$= \frac{15 \times 10^{-5}}{30 \times 10^{-6}} = 5 \text{ volt}$$

$$\text{Now ; } V_L = V_C$$

$$5 = 0.5 \frac{di}{dt}$$

$$\frac{di}{dt} = 10 \text{ A/s}$$

14. Ans (B)

$$P = v \cos \theta T$$

$$T = \frac{2\pi m}{qB}$$

$$P = v \cos \theta \frac{2\pi m}{qB}$$

$$\text{and, } r = \frac{mv \sin \theta}{qB}$$

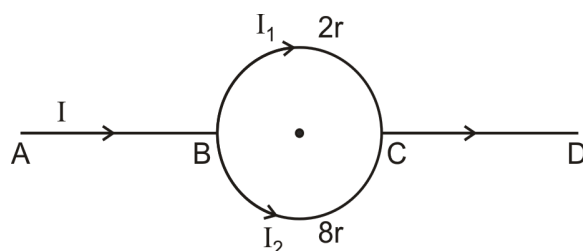
$$\text{given, } P = 2r$$

$$\text{So, } \frac{2\pi m v \cos \theta}{qB} = 2 \left(\frac{mv \sin \theta}{qB} \right)$$

$$\tan \theta = \pi$$

$$\theta = \tan^{-1}(\pi)$$

15. Ans (A)



magnetic field due to wire AB and CD is zero

at center.

$$I_1 = \frac{8r}{2r + 8r} \times I$$

$$\Rightarrow I_1 = \frac{8r}{10r} I \Rightarrow \frac{4}{5} I$$

$$I_2 = \frac{2r}{10r} I \Rightarrow \frac{1}{5} I$$

$$B_1 = \frac{\mu_0 \left(\frac{4I}{5} \right)}{4R} \Rightarrow \frac{\mu_0 I}{5R} \otimes$$

$$B_2 = \frac{\mu_0 \left(\frac{I}{5} \right)}{4R} \Rightarrow \frac{\mu_0 I}{20R} \odot$$

$$B_{\text{net}} = \frac{\mu_0 I}{5R} - \frac{\mu_0 I}{20R}$$

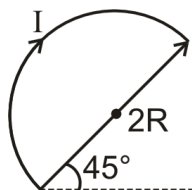
$$\vec{B}_{\text{net}} = \frac{3\mu_0 I}{20R} \otimes$$

16. **Ans (B)**

$$\vec{\ell} = 2R \cos 45^\circ \hat{i} + 2R \sin 45^\circ \hat{j}$$

$$\vec{\ell} = \sqrt{2}R\hat{i} + \sqrt{2}R\hat{j}$$

$$\vec{\ell} = 2\sqrt{2}\hat{i} + 2\sqrt{2}\hat{j}$$



$$\vec{B} = 3\hat{i} + 4\hat{j} - \hat{k}$$

$$\vec{F} = I (\vec{\ell} \times \vec{B})$$

$$\Rightarrow \sqrt{2} (2\sqrt{2}\hat{i} + 2\sqrt{2}\hat{j}) \times (3\hat{i} + 4\hat{j} - \hat{k})$$

$$\vec{F} \Rightarrow -4\hat{i} + 4\hat{j} + 4\hat{k}$$

17. **Ans (C)**

displacement (s) = Area under v - t graph

$$\Rightarrow \frac{1}{2} \times 2 \times 2 \Rightarrow 2\text{m}$$

$$W_{\text{All}} = \Delta KE$$

$$-\mu mgs = 0 - \frac{1}{2}mv^2$$

$$\mu gs = \frac{1}{2}v^2$$

$$\mu \times 10 \times 2 = \frac{1}{2} \times 4$$

$$\mu = 0.1$$

18. **Ans (D)**

$$a = \frac{2mg}{2m + M}$$

$$a = \frac{2(0.1)(10)}{0.2 + 0.5}$$

$$a = \frac{2}{0.7} = \frac{20}{7} \text{ m/s}^2$$

$$\therefore v = u + at$$

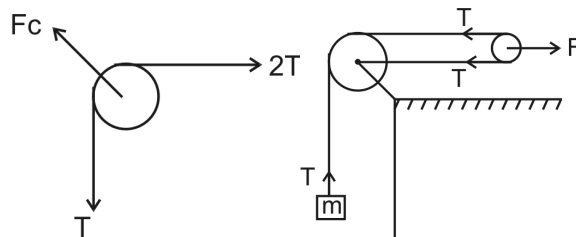
$$0 = 20 - \frac{20}{7}(t)$$

$$\frac{20}{7}(t) = 20$$

$$t = 7$$

19. **Ans (C)**

$$F = 2T$$

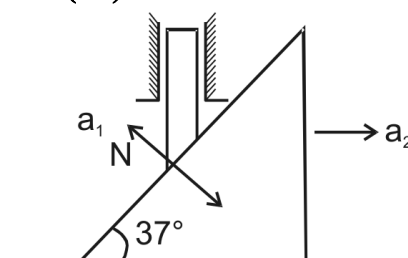


$$F_c = \sqrt{(2T)^2 + T^2}$$

$$F_c = \sqrt{5}T$$

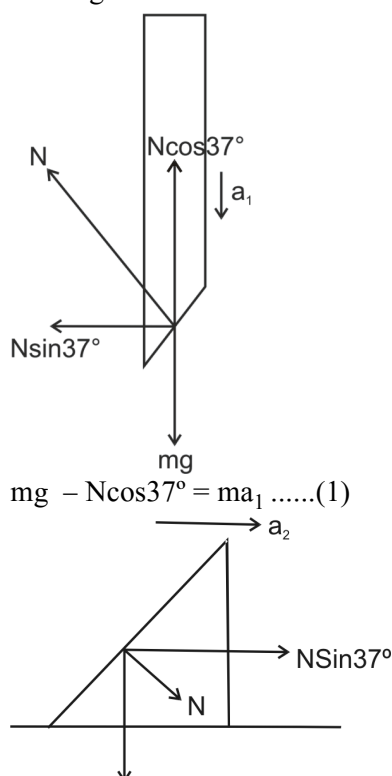
$$\frac{F_c}{F} = \frac{\sqrt{5}T}{2T} = \frac{\sqrt{5}}{2}$$

20. **Ans (A)**



$$m = 5\text{kg}$$

$$M = 9\text{kg}$$



$$mg - N\cos 37^\circ = ma_1 \dots\dots(1)$$

$$N \sin 37^\circ = Ma_2 \dots\dots(2)$$

$$a_1 \cos 37^\circ = a_2 \sin 37^\circ \dots\dots(3)$$

On solving above equations,

$$a_1 = \frac{5g}{21}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (12)**

$$\text{e.m.f.} = \left| -\frac{d\phi}{dt} \right|$$

$$= 108t^2 - 216t$$

$$\text{current} = \frac{108t^2 - 216t}{9}$$

$$i = 12t^2 - 24t$$

$$\frac{di}{dt} = 0 \Rightarrow 24t - 24 = 0$$

$$t = 1 \text{ sec}$$

$$|i_{\max}| \text{ at } t = 1 \text{ sec}$$

$$|i_{\max}| = 12 \text{ A}$$

2. **Ans (2)**

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$

$$\frac{\omega_1}{\omega_2 f_1} + \frac{1}{f_2} = 0$$

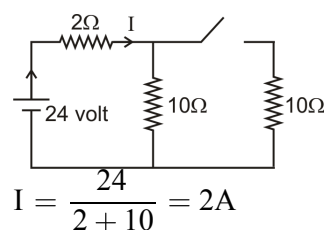
$$\frac{2}{f_1} + \frac{1}{f_2} = 0$$

$$\text{on solving } f_1 = -10 \text{ cm, } f_2 = 5 \text{ cm}$$

3. **Ans (2)**

inductor can be treated as open switch at $t = 0$;

therefore at $t = 0$



4. **Ans (2)**

$$\text{For TIR at face AC, } \mu \geq \csc c \frac{A}{2}$$

5. **Ans (30)**

$$P_{\text{eq}} = 2P_L + P_m$$

$$P_L = (1.5 - 1) \left(\frac{1}{-10} - \frac{1}{20} \right) = \frac{-3}{40}$$

$$P_m = -\frac{1}{f_m} = -\frac{1}{\frac{20}{2}} = -\frac{1}{10}$$

$$P_{\text{eq}} = -\frac{1}{4}$$

$$f_{\text{eq}} = -\frac{1}{P_{\text{eq}}} = 4 \text{ cm}$$

$$v = \frac{uf}{u-f} = \frac{10}{3} \text{ cm}$$

6. **Ans (132)**

$$A = 5 \times 10^{-4} \text{ m}^2$$

$$N = 15$$

$$I = 8 \text{ A}$$

$$B = 22 \text{ T}$$

$$\tau_{\max} = NIAB$$

$$= 15 \times 8 \times 5 \times 10^{-4} \times 22$$

$$\tau_{\max} = 1.32 \text{ N.M}$$

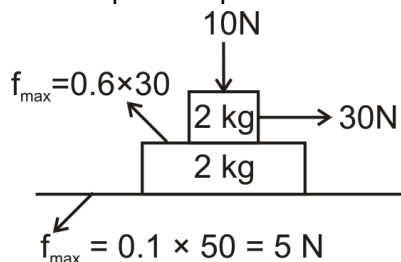
7. **Ans (0)**

Force on closed loop in uniform 'B' = 0.

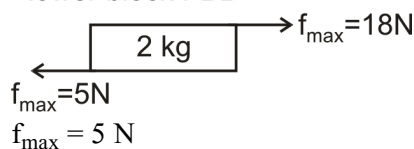
8. **Ans (25)**

If both blocks move together

$$a = \frac{30 - 5}{4} = \frac{25}{4} \text{ m/s}^2$$



lower block FBD

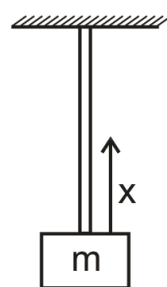


$$\text{Force required} = \frac{25}{4} \times 2 \Rightarrow 12.5 \text{ N}$$

So, sufficient friction is available

$$a = \frac{25}{4} \text{ m/s}^2$$

9. **Ans (7)**

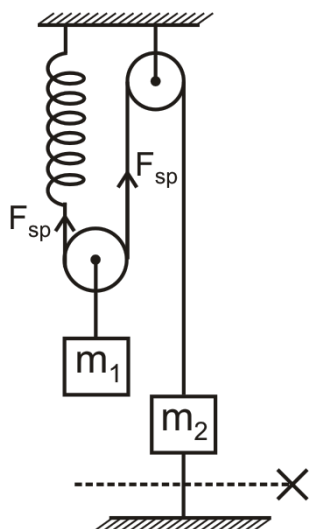


$$T = \left(0.7 + \frac{0.08}{0.2} x \right) 10$$

$$T = 7 + 4x$$

$$\text{so, } \frac{P}{Q} = \frac{7}{4} = 1.75$$

10. Ans (10)



$$2F_{sp} = 200$$

$$F_{sp} = 100$$

F_{BD} of M₂

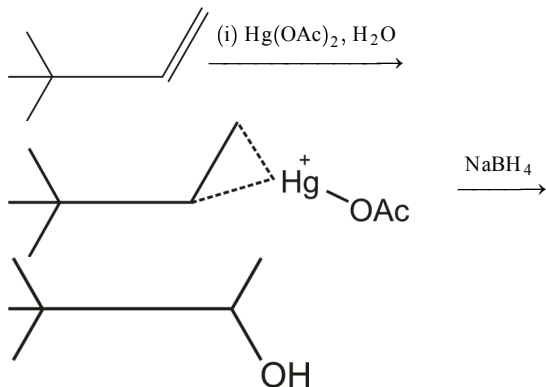
$$\begin{array}{c} \uparrow 100 \\ \boxed{m_2} \\ \downarrow 50 \\ a = \frac{100 - 50}{5} \end{array}$$

$$a = 10 \text{ m/s}^2$$

PART-2 : CHEMISTRY

SECTION-I

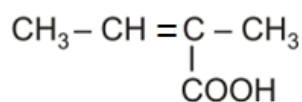
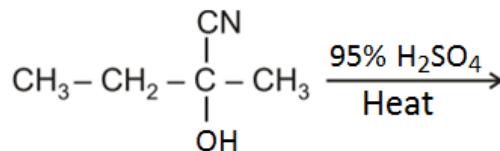
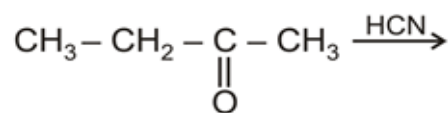
1. Ans (A)



3. Ans (C)

According to concept of DIBAL - H at low temperature reduce ester to aldehyde.

4. Ans (A)



9. Ans (A)

Aromatic compound is more stable than non aromatic and anti aromatic.

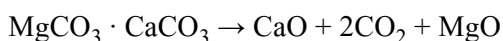
10. Ans (C)

$$\frac{(\text{Probability})_r = 2a_0}{(\text{Probability})_r = a_0} = \frac{4\pi(2a_0)^2 e^{-4} \cdot dr}{4\pi(a_0)^2 e^{-2} \cdot dr} = \frac{4}{e^2}$$

12. Ans (B)

N(SiH₃)₃ molecule is planar due to backbonding (2p → 3d).

13. Ans (A)



$$n_{\text{dolo}} = \frac{18.4}{184} \times 10^3$$

$$n_{\text{CO}_2} = 2 \times 10^2 = 200$$

$$V_{\text{CO}_2} = 200 \times 24.63 = 4926$$

14. Ans (C)

$$X = E_0/n^2 \quad Y = E_0/(n+4)^2$$

$$\frac{X}{Y} = \left(\frac{n+4}{n} \right)^2$$

$$\Rightarrow \sqrt{\frac{X}{Y}} = 1 + \frac{4}{n}$$

15. Ans (C)

Azurite is a ore of copper.

18. Ans (C)

	PbSO ₄	ZnS	AgBr	CuCO ₃
K _{sp}	2×10^{-9}	1×10^{-22}	4×10^{-13}	1×10^{-8}
S(in Mole/L)	447×10^{-5}	1×10^{-11}	632×10^{-7}	1×10^{-4}
S(in g/ml)	136×10^{-3}	9.7×10^{-10}	1.18×10^{-4}	123×10^{-3}

19. Ans (D)

$$p^H = 13$$

$$[H^+] = 10^{-13}$$

$$\text{In } 1 \text{ L} = 10^{-13} \text{ mole of } H^+$$

$$\text{In } 1 \text{ ml} = 10^{-16} \text{ mole of } H^+$$

$$\text{No of } H^+ = 10^{-16} \times 6.022 \times 10^{23}$$

$$= 6.022 \times 10^7$$

20. Ans (A)

For Li²⁺ ion,

$$\Delta E = 13.6 \times (3)^2 \left(\frac{1}{3^2} - \frac{1}{6^2} \right) \text{ eV} = 13.6 \times 3/4 \text{ eV}$$

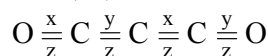
For H-atom (n = 2 to 1)

$$\Delta E = 13.6 \times (1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \text{ eV} = 13.6 \times 3/4 \text{ eV}$$

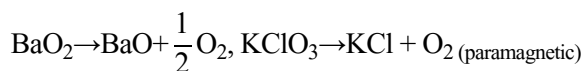
PART-2 : CHEMISTRY

SECTION-II

1. Ans (2)



2. Ans (6)



3. Ans (917)

$$\Delta E = \Delta E_1 + \Delta E_2$$

$$= 1310 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) \times \frac{45}{100} + 1310 \times \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \times \frac{40}{100}$$

$$= 917 \text{ kJ}$$

6. Ans (7)

Active methylene group will give positive iodoform test.

7. Ans (2)

More stable the conjugate base more the acidic nature of its conjugate.

(iii), (v)

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A)

$$I = \int \frac{(f(x) - f'(x)) e^x}{(e^x + f(x))^2} dx = \int \frac{e^{-x} (f(x) - f'(x))}{(1 + e^{-x} f(x))^2} dx$$

$$\text{Put } \frac{1}{1 + e^{-x} f(x)} = t$$

$$\Rightarrow \frac{-e^{-x} f(x) + f(x) e^{-x}}{(1 + e^{-x} f(x))^2} dx = dt$$

$$\Rightarrow \frac{e^{-x} (f(x) - f'(x))}{(1 + e^{-x} f(x))^2} dx = dt$$

$$\therefore I = t + C = \frac{1}{1 + e^{-x} f(x)} + C$$

2. Ans (A)

$$\text{Let } I = \frac{1}{2} \int e^{\sin^2 x} (1 + \cos^2 x) \sin 2x dx$$

$$\text{Put, } \sin^2 x = t \Rightarrow \sin 2x dx = dt,$$

$$\text{So, } I = \frac{1}{2} \int e^t (1 + 1 - t) dt = \frac{1}{2} \int \underbrace{e^t}_I \underbrace{(2-t)}_I dt$$

$$= \frac{1}{2} \left[(2-t) e^t + \int e^t dt \right] = \frac{1}{2} [(2-t) e^t + e^t] + C$$

$$= \frac{1}{2} (3-t) e^t + C = \frac{1}{2} (3 - \sin^2 x) e^{\sin^2 x} + C$$

3. **Ans (B)**

$$I_n = \int \cot^n x dx = \int \cot^{n-2} \cdot x \cdot (\operatorname{cosec}^2 x - 1) dx$$

$$\text{Let } u = \cot(x)$$

$$I_n = -\frac{u^{n-1}}{n-1} - I_{n-2}$$

$$\text{or } I_n + I_{n-2} = -\frac{u^{n-1}}{n-1} \quad (\text{Put } n = 2, 3, 4, \dots, 10)$$

$$I_2 + I_0 = -\frac{u}{1}$$

$$I_3 + I_1 = -\frac{u^2}{2}$$

$$I_4 + I_2 = -\frac{u^3}{3}$$

.....

$$I_{10} + I_8 = -\frac{u^9}{9}$$

$$\text{adding } I_0 + I_1 + 2(I_2 + I_3 + \dots + I_8) + I_9 + I_{10}$$

$$= -\left(u + \frac{u^2}{2} + \dots + \frac{u^9}{9}\right)$$

$$\Rightarrow (B)$$

4. **Ans (D)**

$$I = \int \frac{3e^x + 5e^{-x}}{4e^x - 5e^{-x}} dx = \int \frac{3e^{2x} + 5}{4e^{2x} - 5} dx$$

$$\text{Let } 3e^{2x} + 5 = A(4e^{2x} - 5) + B(8e^{2x})$$

$$\therefore 4A + 8B = 3 \quad \dots\dots(1)$$

$$\text{and } -5A = 5 \Rightarrow A = -1 \quad \dots\dots(2)$$

$$\therefore B = \frac{7}{8}$$

$$\therefore I = Ax + B \ln(4e^{2x} - 5) + C \quad \text{when } A = -1$$

$$\text{and } B = \frac{7}{8}$$

$$A + B = \frac{-1}{8} \Rightarrow [A + B] = -1$$

5. **Ans (C)**

$$I = \lim_{k \rightarrow 0} \frac{\int_0^k (1 + \sin 2x)^{\frac{1}{x}} dx}{k}$$

differentiating, w.r.t. k using L'Hospital's rule we get

$$I = \lim_{k \rightarrow 0} (1 + \sin 2k)^{\frac{1}{k}} = e^{\lim_{k \rightarrow 0} \frac{1}{k} (\sin 2k)} = e^2$$

6. **Ans (A)**

$$a_n = \int_0^{\pi/2} (1 - \sin t)^n \sin 2t dt$$

$$\text{Let } 1 - \sin t = u$$

$$\Rightarrow -\cos t dt = du$$

$$a_n = 2 \int_0^1 u^n (1 - u) du = 2 \left(\int_0^1 u^n du - \int_0^1 u^{n+1} du \right)$$

$$= 2 \left(\frac{1}{n+1} - \frac{1}{n+2} \right)$$

$$\text{hence } \frac{a_n}{n} = 2 \left(\frac{1}{n(n+1)} - \frac{1}{n(n+2)} \right)$$

$$\lim_{n \rightarrow \infty} \sum_{n=1}^n \frac{a_n}{n}$$

$$= 2 \left(\sum \left(\frac{1}{n} - \frac{1}{n+1} \right) - \frac{1}{2} \sum \left(\frac{1}{n} - \frac{1}{n+2} \right) \right)$$

$$= 2 \sum_{n=1}^n \left(\frac{1}{n} - \frac{1}{n+1} \right) - \sum_{n=1}^n \left(\frac{1}{n} - \frac{1}{n+2} \right)$$

$$= 2(1) - \left[\left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \dots \right]$$

$$= 2 - \frac{3}{2} = \frac{1}{2}$$

7. **Ans (A)**

$$\text{Let } \int_0^x f(t) dt = T(x)$$

$$\Rightarrow T'(x) = f(x)$$

$\therefore$ On differentiating b.t.s. w.r.t. x, we get

$$f(x) = T'(x)$$

Hence

$$G(x) = \int e^x \left(\int_0^x f(t) dt + f(x) \right) dx$$

$$= \int e^x (T(x) + T'(x)) dx = e^x T(x) + C$$

$$\Rightarrow G(x) = e^x \int_0^x f(t) dt + C$$

Now on differentiating

$$G'(x) = e^x \int_0^x f(t) dt + e^x f(x) \Rightarrow G'(0) = f(0)$$

$$= 1$$

8. Ans (C)

$$T_r = \frac{1}{\sqrt{\frac{r}{n}} \cdot n \left(3\sqrt{\frac{r}{n}} + 4 \right)^2}$$

$$S = \frac{1}{n} \sum_{i=1}^{4n} \frac{1}{\left(3\sqrt{\frac{r}{n}} + 4 \right)^2 \cdot \sqrt{\frac{r}{n}}} = \int_0^4 \frac{dx}{\sqrt{x} (3\sqrt{x} + 4)^2}$$

$$\text{put } 3\sqrt{x} + 4 = t$$

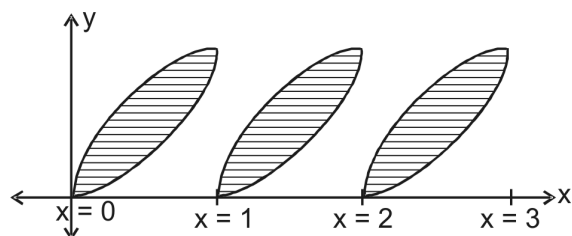
$$\Rightarrow \frac{3}{2} \frac{1}{\sqrt{x}} dx = dt$$

$$= \frac{2}{3} \int_4^{10} \frac{dt}{t^2} = \frac{2}{3} \left[\frac{1}{t} \right]_4^{10} = \frac{2}{3} \left[\frac{1}{4} - \frac{1}{10} \right]$$

$$= \frac{2}{3} \cdot \frac{6}{40} = \frac{1}{10}$$

9. Ans (C)

Both functions are periodic with period 1



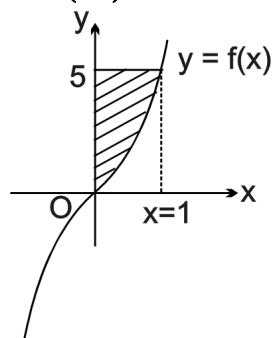
Hence area

$$= 10 \int_0^1 (\sqrt{x} - x^2) dx$$

$$= 10 \left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{3} \right)_0^1$$

$$= 10 \left(\frac{2}{3} - \frac{1}{3} \right) = \frac{10}{3}$$

10. Ans (D)



Note for inverse function y axis will be the x axis and x axis will be the y axis

$$\text{required area} = \text{Area of rectangle} - \int_0^1 f(x) dx$$

$$= 5 - \int_0^1 (3x^3 + 2x) dx$$

$$= 5 - \left(\frac{3}{4} + 1 \right) = 3 \frac{1}{4} = \frac{13}{4}$$

11. Ans (D)

$$y \sin 2x - \cos x + (1 + \sin^2 x) \frac{dy}{dx} = 0$$

where $y = f(x)$

$$\frac{dy}{dx} + \left(\frac{\sin 2x}{1 + \sin^2 x} \right) y = \frac{\cos x}{1 + \sin^2 x}$$

$$\text{I.F.} = e^{\int \frac{\sin 2x}{1 + \sin^2 x} dx} = e^{\int \frac{dt}{t}} = e^{\ln(1 + \sin^2 x)}$$

$$= 1 + \sin^2 x \quad (\text{by putting } 1 + \sin^2 x = t)$$

$$y(1 + \sin^2 x) = \int \cos x dx$$

$$y(1 + \sin^2 x) = \sin x + C; (y(0) = 0) \Rightarrow C = 0$$

$$\text{hence, } y = \frac{\sin x}{1 + \sin^2 x}$$

$$y \left(\frac{\pi}{6} \right) = \frac{2}{5}$$

12. Ans (C)

Given,

$$\frac{xdy - ydx}{y^2} = dy \Rightarrow d \left(\frac{x}{y} \right) = -dy$$

$$\Rightarrow \frac{x}{y} = -y + C$$

$$\text{As, } y(1) = 1 \Rightarrow C = 2 \Rightarrow \frac{x}{y} + y = 2$$

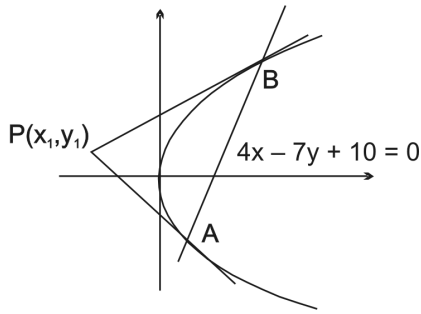
Put $x = -3$, we get

$$-3 + y^2 = 2y \Rightarrow (y + 1)(y - 3) = 0$$

As $y > 0$

$$\text{So, } y(x = -3) = 3$$

13. Ans (C)



C.O.C. of $P(x_1, y_1)$

w.r.t. $y^2 = 4x$ is

$$yy_1 = 2(x + x_1) \quad \dots(1)$$

compare with

$$4x - 7y + 10 = 0 \quad \dots(2)$$

$$\text{to get } (x_1, y_1) = \left(\frac{5}{2}, \frac{7}{2}\right)$$

14. Ans (C)

$$A = \frac{(y_1^2 - 4ax_1)^{3/2}}{2a}; x_1 = 4; y_1 = 6; a = 2$$

15. Ans (D)

By using condition of tangency, we get $4h^2 = 3k^2 + 2$

$\therefore$ Locus of $P(h, k)$ is $4x^2 - 3y^2 = 2$ (which is hyperbola.)

$$\text{Hence } e^2 = 1 + \frac{4}{3} \Rightarrow e = \sqrt{\frac{7}{3}}$$

16. Ans (A)

Chord of the hyperbola $x^2 - y^2 = a^2$ with

middle point (h, k) , is $hx - ky = h^2 - k^2$

$$\Rightarrow y = \frac{h}{k}x + \frac{k^2 - h^2}{k} \quad \dots\dots(1)$$

$$\left(m = \frac{h}{k}; c = \frac{k^2 - h^2}{k}\right)$$

As (1) touches $y^2 = 4ax$, condition of tangency gives

$$c = \frac{a}{m} \Rightarrow \frac{k^2 - h^2}{k} = \frac{ak}{h}$$

$$\Rightarrow x(y^2 - x^2) = 3y^2 \Rightarrow x^3 = y^2(x - 3)$$

17. Ans (A)

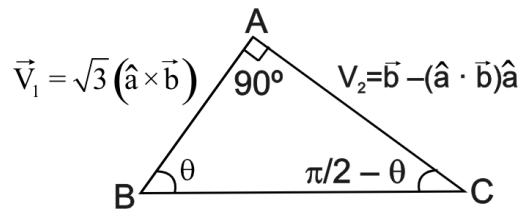
$$\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} = \begin{vmatrix} t & -3 & 2t \\ 1 & -2 & 2 \\ 3 & t & -1 \end{vmatrix} = t(2 - 2t) + 3(-7) + 2t(t + 6) = 7(2t - 3)$$

$$+ 2t(t + 6) = 7(2t - 3)$$

$$\int_1^2 \vec{a} \cdot (\vec{b} \times \vec{c}) dt = 7 \int_1^2 (2t - 3) dt = 7$$

$$[t^2 - 3t]_1^2 = 7[(4 - 6) - (1 - 3)] = 0$$

18. Ans (B)

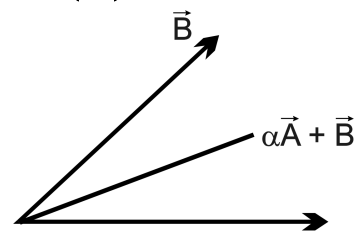


Consider $\vec{V}_1 \cdot \vec{V}_2 = 0 \Rightarrow A = 90^\circ$

$$\text{Using sine law, } \frac{|\vec{b} - (\hat{a} \cdot \vec{b}) \hat{a}|}{\sin \theta} = \frac{\sqrt{3} |\hat{a} \times \vec{b}|}{\cos \theta}$$

$$\tan \theta = \frac{1}{\sqrt{3}} \frac{|\vec{b} - (\hat{a} \cdot \vec{b}) \hat{a}|}{|\vec{a} \times \vec{b}|} = \frac{1}{\sqrt{3}} \frac{|(\hat{a} \times \vec{b}) \times \hat{a}|}{|\vec{a} \times \vec{b}|} = \frac{1}{\sqrt{3}} \frac{|\hat{a} \times \vec{b}| |\vec{a}| \sin 90^\circ}{|\vec{a} \times \vec{b}|} = \frac{1}{\sqrt{3}} \Rightarrow \theta = \frac{\pi}{6}$$

19. Ans (B)



$$\vec{A} \cdot (\alpha \vec{A} + \vec{B}) = \vec{B} \cdot (\alpha \vec{A} + \vec{B})$$

$$\Rightarrow \alpha + \vec{A} \cdot \vec{B} = \alpha \vec{A} \cdot \vec{B} + 1$$

$$(\vec{A} \cdot \vec{B})(1 - \alpha) = (1 - \alpha)$$

$$(1 - \alpha)[\vec{A} \cdot \vec{B} - 1] = 0$$

$$\text{if } \vec{A} \cdot \vec{B} = 1$$

$$\Rightarrow \cos \theta = 1$$

$$\Rightarrow \theta = 0 \text{ which is not possible}$$

$$\Rightarrow \alpha = 1$$

20. Ans (A)

$$|\vec{u}|$$

$$= \sqrt{16e^{8t} \cos^2 3t - 24e^{8t} \cos 3t \sin 3t + 9e^{8t} \sin^2 3t + 16e^{8t} \sin^2 3t + 24e^{8t} \cos 3t \sin 3t + 9e^{8t} \cos^2 3t}$$

$$= \sqrt{16e^{8t} + 9e^{8t}} = 5e^{4t}$$

So $f(t)$ is strictly increasing so maximum value at $t = 5$ and minimum at $t = 0$,

$$\therefore \frac{f(5)}{f(0)} = \frac{5e^{20}}{5} = e^{20}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (12)

$$3f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} - 10 \quad \dots(1)$$

replace $x \rightarrow \frac{1}{x}$ to get

$$3f\left(\frac{1}{x}\right) + 4f(x) = x - 10 \quad \dots(2)$$

Solving (1) and (2) we get

$$f(x) = \frac{4x}{7} - \frac{3}{7x} - \frac{10}{7}$$

$$\int_2^3 f(x) dx = \int_2^3 \left(\frac{4x}{7} - \frac{3}{7x} - \frac{10}{7} \right) dx$$

$$= \frac{3}{7} \ln\left(\frac{2}{3}\right) = \frac{a}{b} \ln\left(\frac{c}{a}\right)$$

Hence $a = 3$, $b = 7$, $c = 2$

$$\therefore (a + b + c) = 12.$$

2. Ans (2010)

L.H.S.

$$\int \frac{(\sin x)^{2008} - (\cos x)^{2008}}{(\sin x)^{2008} \left(\frac{\sin x}{\cos x} + \left(\frac{\cos x}{\sin x} \right)^{2009} \right)} dx$$

$$= \int \frac{\sin x \cos x \left((\sin x)^{2008} - (\cos x)^{2008} \right)}{(\sin x)^{2010} + (\cos x)^{2010}} dx$$

$$= \int \frac{\left((\sin x)^{2009} \cos x - (\cos x)^{2009} \sin x \right)}{(\sin x)^{2010} + (\cos x)^{2010}} dx$$

$$\text{put } (\sin x)^{2010} + (\cos x)^{2010} = t$$

$$= \frac{1}{2010} \int \frac{dt}{t}$$

$$= \frac{1}{2010} \ln |(\sin x)^{2010} + (\cos x)^{2010}| + c$$

$$\Rightarrow k = 2010$$

3. Ans (2014)

$$I = \int_0^{\pi/2} (\cos x)^{2011} (\sin 2012x + x) dx$$

$$I = \int_0^{\pi/2} (\cos x)^{2011} (\sin 2012x \cos x + \cos 2012x \sin x) dx$$

$$I = \int_0^{\pi/2} \left(\sin(2012x) (\cos x)^{2012} + (\cos 2012x) \sin x (\cos x)^{2011} \right) dx$$

$$I = \frac{-1}{2012} \int_0^{\pi/2} \frac{d}{dx} \left(\cos 2012x \cdot (\cos x)^{2012} \right) dx$$

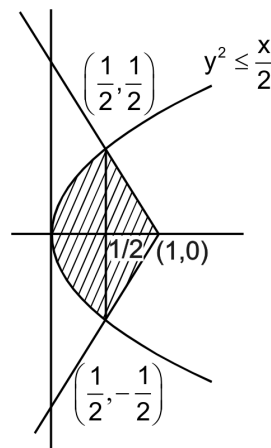
$$I = \frac{-1}{2012} \left((\cos 2012x) (\cos x)^{2012} \right) \Big|_0^{\pi/2}$$

$$= \frac{-1}{2012} (-1) = \frac{1}{2012} = \frac{a}{b}$$

$$\therefore a = 1; b = 2012$$

$$(2a + b) = 2014$$

4. Ans (7)



Solving $x = 2y^2$ and $|y| = 1 - x$, we get

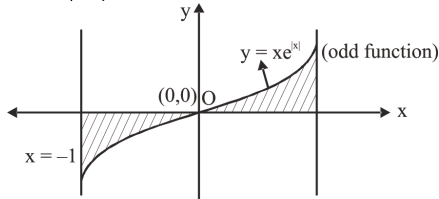
$$x = \frac{1}{2} \quad \text{and} \quad y = \frac{1}{2} \quad \text{or} \quad x = \frac{1}{2} \quad \text{and} \quad y = -\frac{1}{2}$$

$$\text{hence, } A = \frac{2}{\sqrt{2}} \int_0^{1/2} \sqrt{x} dx + \frac{1}{2} \cdot 1 \cdot \frac{1}{2} = \frac{1}{3} + \frac{1}{4}$$

$$= \frac{7}{12}$$

$$\therefore 12A = 7$$

5. Ans (2)



$$f(x) = xe^{|x|} = \begin{cases} xe^{-x}, & x < 0 \\ xe^x, & x \geq 0 \end{cases}$$

Note : $f(x)$ is an increasing function on $\mathbb{R}$.

$$\begin{aligned} \text{So, required area} &= 2 \int_0^1 x e^x dx = 2(x e^x - e^x) \Big|_0^1 \\ &= 2((e - e) - (0 - 1)) = 2 \end{aligned}$$

6. Ans (9)

$$\begin{aligned} \int_0^x f(t) dt + \int_0^x (x-t) f(t) dt &= e^{-x} - 1 \\ x \int_0^x f(t) dt + \int_0^x f(t) (1-t) dt &= e^{-x} - 1 \end{aligned}$$

differentiate w.r.t. x

$$x \cdot f(x) + \int_0^x f(t) dt + f(x) (1-x) = -e^{-x}.$$

again differentiate w.r.t. x

$$\begin{aligned} x f'(x) + f(x) + f(x) + f'(x) - x f'(x) - f(x) &= e^{-x} \\ f'(x) + f(x) &= e^{-x} \end{aligned}$$

$$\text{I.F.} = e^x \Rightarrow y \cdot ex = \int dx$$

$$f(x) e^x = x + c ; f(0) = -1$$

$$\therefore f(x) e^x = x - 1$$

$$x = 10$$

$$\Rightarrow e^{10} f(10) = 9$$

7. Ans (10)

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 ; \text{ Here, } m = -1 \quad (\text{Given})$$

$$y = mx \pm \sqrt{16m^2 + 9} \quad (\text{equation of tangent})$$

$$\Rightarrow y = -x \pm 5$$

$$x + y \pm 5 = 0$$

as two lines are parallel so that the circle will touch both hence distance between them is

$$\text{diameter of circle i.e. } \frac{10}{\sqrt{2}}.$$

$$\therefore \text{Diameter of director circle} = \frac{10}{\sqrt{2}} \times \sqrt{2} = 10$$

8. Ans (8)

Let variable chord be $x \cos \alpha + y \sin \alpha = p$

homogenizing the hyperbola

$$\frac{x^2}{4} - \frac{y^2}{8} = \frac{(x \cos \alpha + y \sin \alpha)^2}{p^2}$$

Now, coefficient of x^2 + coefficient of y^2 = 0

$$\frac{\cos^2 \alpha}{p^2} - \frac{1}{4} + \frac{\sin^2 \alpha}{p^2} + \frac{1}{8} = 0$$

$$\frac{1}{p^2} = \frac{1}{8}$$

$$\Rightarrow p^2 = 8$$

$$\therefore \text{required } R^2 = 8$$

9. Ans (15)

$$\text{Given } \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = V_1$$

$$\begin{bmatrix} \vec{p} & \vec{q} & \vec{r} \end{bmatrix} = \begin{vmatrix} 1 & 1 & -2 \\ 3 & -2 & +1 \\ 1 & -4 & 2 \end{vmatrix} \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}$$

$$= [1(-4+4) - 1(6-1) - 2(-12+2)] V_1$$

$$V_2 = (-5+20) V_1 = 15V_1$$

$$\Rightarrow K = 15$$

10. Ans (5)

$$(\vec{p} \cdot \vec{r}) \vec{q} - (\vec{q} \cdot \vec{r}) \vec{p} + (\vec{q} \cdot \vec{r}) \vec{q} = (x^2 + y^2) \vec{q} + (14 - 4x - 6y) \vec{p}$$

$$\therefore \vec{p} \cdot \vec{r} + \vec{q} \cdot \vec{r} = x^2 + y^2 \quad \dots(1)$$

$$\text{and } -(\vec{q} \cdot \vec{r}) = 14 - 4x - 6y \quad \dots(2)$$

From (1) + (2)

$$\vec{p} \cdot \vec{r} = x^2 + y^2 - 4x - 6y + 14 \quad \dots(3)$$

$$(\vec{r} \cdot \vec{r}) \vec{p} = \vec{r}$$

Taking dot product with $\vec{r}$, we get

$$(\vec{r} \cdot \vec{r}) (\vec{p} \cdot \vec{r}) = \vec{r} \cdot \vec{r} \Rightarrow \vec{p} \cdot \vec{r} = 1$$

$\therefore$ from (3)

$$x^2 + y^2 - 4x - 6y + 14 = 1$$

$$\Rightarrow (x-2)^2 + (y-3)^2 = 0$$

$$\Rightarrow x = 2 \text{ \& } y = 3$$

$$\text{Hence } (x+y) = 5$$